

Household Income Polarization in Italy: A Historical Comparison from 2000 to 2020

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Abstract

Using the relative distribution method and applying a kernel estimator suitable for variables with left-bounded support, the objective of this work is to search for possible traces of polarization in the distribution of disposable income and in the distributions of different income sources, in household terms, in Italy from 2000 to 2020.

We show that in many of the cases considered, apart from a few cases of depolarization, polarization manifests as bipolarization or multipolarization, reflected in the disappearance or fragmentation of the middle class or particular groups. Furthermore, with particular reference to the Italian case, we provide some possible effects arising from the presence of the polarization phenomenon.

1 Introduction

Given its now-recognized complementarity with studies on inequality, the study of polarization is of considerable importance in income analysis. This phenomenon has been studied in various contexts, considering different years, different indicators, and applying different methodologies. Nevertheless, on the one hand, there is currently no generally accepted approach to study it, and on the other hand, it is a phenomenon still poorly monitored in some countries. Furthermore, in our opinion, only some of its potential effects on society and the economy are known.

Currently, to our knowledge, empirical contributions regarding the investigation of income polarization in Italy are extremely limited, both because they consider datasets with few years and because they do not consider the most recent years. Our contribution aims to fill this gap.

Using the relative distribution method, proposed by Handcock & Morris (1998, 1999, 2002, 2006), and the density function estimator proposed by Botev, Grotowski & Kroese (2010), which is particularly suitable for the data we consider, we show the traces of income polarization and its evolution in Italy between 2000 and 2020.

We find that in many cases, and in various forms, the phenomenon of polarization has been present in Italy in the years considered. In particular, we find cases of: bipolarization (understood as increases in density not necessarily in extreme quantile intervals of the distribution), unbalanced bipolarization (towards the left or right), tri-polarization, quadri-polarization, depolarization. Where cases of bipolarization and multipolarization are not always associated with a total or prevalent disappearance of the middle class.

median, or towards a more or less central interval of the distribution, and a consequent decrease in density at more or less extreme intervals of the distribution (in terms of overall relative distribution and/or shape effect).

Regarding the second group of outputs, the indicators, they are the median relative polarization index (Equation 3), lower relative polarization index (Equation 4), and upper relative polarization index (Equation 5):

$$MRP(F; F_0) = 4 \int_0^1 \left| r - \frac{1}{2} \right| g_0^A(x) dr - 1 \quad (3)$$

$$LRP(F; F_0) = 8 \int_0^{1/2} \left| r - \frac{1}{2} \right| g_m(x) dr - 1 \quad (4)$$

$$URP(F; F_0) = 8 \int_{1/2}^1 \left| r - \frac{1}{2} \right| g_m(x) dr - 1 \quad (5)$$

Where g_0^A is the mean absolute deviation around the median of the location-matched relative distribution, $|r - \frac{1}{2}|$ is a weight such that the indicator varies between -1 and 1, making the deviation in the tails more evident.

All the indicators presented above have similar interpretations: positive values indicate greater polarization, and therefore a greater concentration of density in the tails of the distribution; negative values indicate the convergence of density towards the center (depolarization); a value of zero indicates no difference in shape between the distributions.

It is worth specifying that the relative distribution method provides graphical outputs useful for studying polarization both in terms of bipolarization and multipolarization. However, for the elements that we consider symptomatic of polarization and given our broad interpretation which, while not distorting the essence of the phenomenon, is not strictly linked to bipolarization, the indicators do not provide particular information and are sometimes contradictory to what emerges from the graphs. This is because they are constructed to be particularly sensitive to cases of bipolarization (and, more precisely, to cases of bipolarization where density is concentrated in the tails of the distribution) but not very sensitive to cases where multiple distinct and separate peaks emerge along the entire function (multipolarization). In view of this consideration, we deem the polarization indicators proposed within this method to be unreliable in our case and will focus on a coordinated analysis of the overall relative distribution and shape effect functions.

3.2 The BGK Estimator

Botev, Grotowski & Kroese (2010) proposed a kernel estimator that performs particularly well even in the case of data with limited support, hence our choice to use it to estimate the density functions (of disposable income and income sources) to be included in the relative distribution.

More precisely, the authors propose an adaptive kernel estimator, which is based on Gaussian kernel estimation, linear diffusion processes, and adaptive smoothing, taking into account information from the estimated pilot density. Furthermore, they provide a plug-in bandwidth selection method that does

not require normality assumptions for the data and whose choice does not require particular computational resources.

Given the mathematical complexity in the work of the three authors, reporting even some parts of which would lead to a considerable lengthening of this section, we provide below only some elements to highlight which characteristics of the estimator in question make it particularly suitable for our purposes⁷.

Given a variable X with support $\mathcal{X}$, the proposed estimator for estimating the density function of X is

$$g(x; t) = \frac{1}{N} \sum_{i=1}^N k(x, X_i; t) \quad \forall y \in \mathcal{X} \quad (6)$$

where for each fixed $y \in \mathcal{X}$ the diffusion kernel k must satisfy

$$\begin{cases} \frac{\partial}{\partial t} k(x, y; t) = Lk(x, y; t) & \text{with } x \in \mathcal{X}, t > 0 \\ k(x, y; 0) = \delta(x - y) & \text{with } x \in \mathcal{X} \end{cases} \quad (7)$$

and where for each fixed $x \in \mathcal{X}$ the diffusion kernel k must satisfy

$$\begin{cases} \frac{\partial}{\partial t} k(x, y; t) = L^*k(x, y; t) & \text{with } y \in \mathcal{X}, t > 0 \\ k(x, y; 0) = \delta(x - y) & \text{with } y \in \mathcal{X} \end{cases} \quad (8)$$

Where L^* is the adjoint operator of L and is also the infinitesimal generator of the diffusion process assumed by the authors, k is the kernel function and $\sqrt{t}$ is the bandwidth.

If $\mathcal{X}$ has limitations, then a further condition is added (Neumann boundary condition) according to which

$$\frac{\partial}{\partial t} \left(\frac{k(x, y; t)}{p(x)} \right) \Big|_{x \in \partial \mathcal{X}} = 0 \quad \forall t > 0 \quad (9)$$

Finally, in the course of their work, the authors demonstrate, among other things, that the estimator is consistent in the case of support $[0, +\infty]$.

3.3 Data and Software

In this study, we considered data relating to annual household income exclusively from the "Survey on Household Income and Wealth" (Banca d'Italia) and, more precisely, those relating to the years 2000-2020 extractable from the "comp", "peso", and "rfam" datasets of the historical archive. These data are freely accessible and downloadable from the Bank of Italy's website.

Compared to the original data, we applied three modifications. In fact, we took into account the modified OECD scale, the sample weights for historical comparison, and we eliminated observations that had a null or negative value.

First, according to the OECD equivalence scale, a value of 1 is assigned to the first adult individual in the household, 0.5 for each other adult or child over 14 years of age, and 0.3 for each child under 14

⁷For the choice and estimation of the optimal bandwidth, as well as for further details and properties, see the work of Botev, Grotowski & Kroese (2010).

sources of income, is that designing some public policies (for example, to support the self-employed or involving pensions, or that affect economic support for families) could become increasingly complicated if there is no prevalent group (middle class) that potentially benefits from the effects of such a policy. In practice, it would be easier for governments to meet the needs of a more or less broad middle class rather than the needs of two distinct and opposing groups that have different income characteristics and could have very different needs and preferences.

We note two important facts.

First, the said possible consequences of bipolarization are certainly not exhaustive. In fact, by analyzing the relationships between some variables and observing some situations present in the Italian context, further cases for reflection could be found.

Second, above we referred to the case of bipolarization both because it is the main case in which the central quantiles can undergo relevant reductions in density, and because it is the simplest case to analyze in an abstract exercise. Nevertheless, it is also conceivable to be able to extend the previous reflections to more general cases of polarization where the support of a distribution is fragmented into several intervals within which the density increases, thus generating different groups, not necessarily extreme and very distant, distinct and, probably, with different characteristics. In this perspective, for example, one could think that polarization processes could fragment the middle class, but also the groups of the richest and the poorest; or still, generate population groups with particular consumption habits; or further, groups with needs so diversified as to make the design of public policies very complex.

As highlighted by several authors, including Franzini (2019) and Banerjee & Duflo (2008), we also argue here that polarization in Italy could be a destabilizing and harmful phenomenon both at an economic and social level.

5 Conclusions

With this work, we show that, given the tools used, income polarization in Italy, between the years 2000 and 2020, has been an almost ever-present phenomenon with various characteristics depending on the source of income and the period considered.

We have found and documented cases of: bipolarization, understood as increases in density not necessarily in extreme quantile intervals of the distribution, including cases of bipolarization skewed to the left and to the right; cases of multipolarization, more precisely of tri-polarization and quadri-polarization; cases, very few, of depolarization.

The detected cases have different characteristics from each other and it seems that cases of bipolarization and multipolarization are not always associated with a total or prevalent disappearance of the middle class. In fact, increases in density are not necessarily observed in the most extreme quantiles of the distribution and reductions are not necessarily observed at the most central quantiles, but rather both seem to involve fractions of the distributions of disposable income and of the different sources of income in household terms in a relevant way.

The possible implications generated by the presence of this phenomenon in Italy are many and di-

verse. They can involve aspects of aggregate demand, the supply of services, as well as the ability of governments to implement expansionary fiscal policies and the degree of difficulty in designing policies.

The need to monitor, study, and understand this phenomenon is crucial both because of its link with inequality and for the possible consequences in the economic and social fields.

The possible future developments in the study of income polarization are, in our opinion, very numerous. To date, there is no generally shared definition of polarization. As far as we know, there are no works mainly oriented towards economic policy that focus exclusively on preventing or combating cases of polarization, nor on the policy tools potentially useful in contexts of income polarization. It is still little known what triggers and what the presence of this phenomenon implies. A further possible development concerns the determination of indicators and thresholds of polarization that are objectively classifiable as high and low. Again, it is not known whether there is a "physiological" level of polarization within an economy.

6 Appendix 1: Histograms

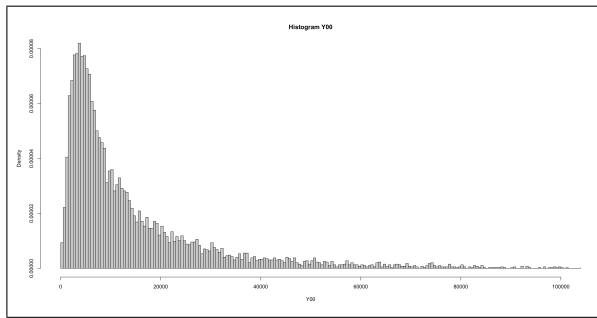


Figure 1: Histogram Y00

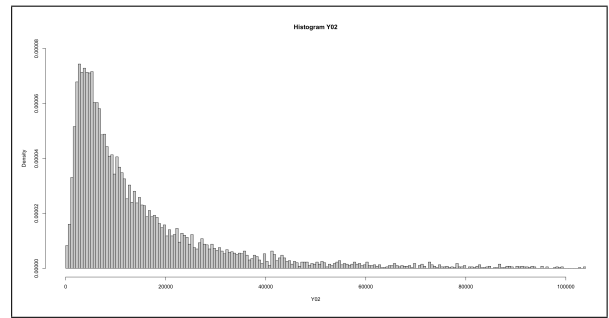


Figure 2: Histogram Y02

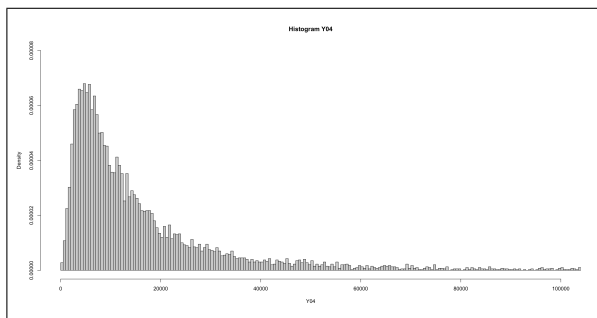


Figure 3: Histogram Y04

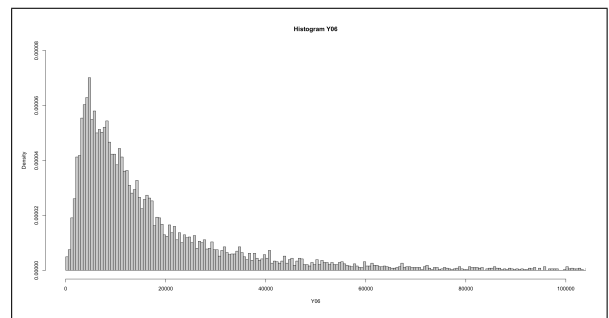


Figure 4: Histogram Y06

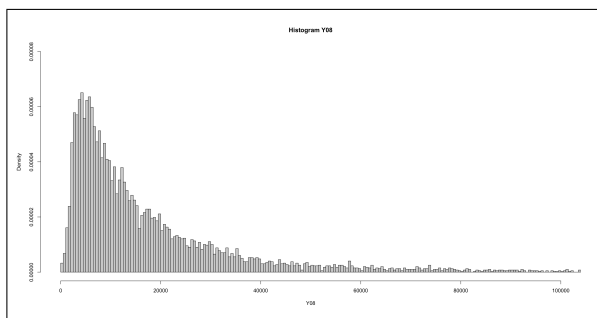


Figure 5: Histogram Y08

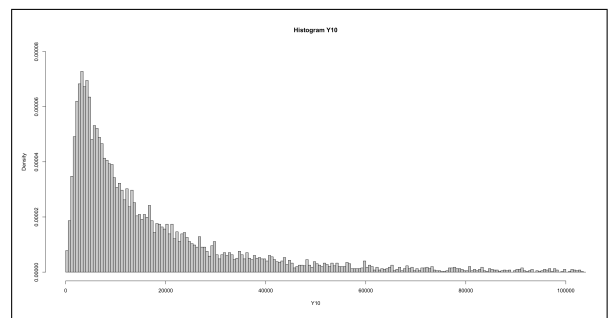


Figure 6: Histogram Y10

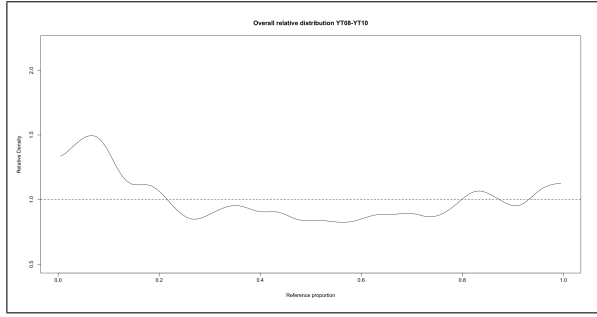


Figure 153: Overall relative distribution YT08-YT10

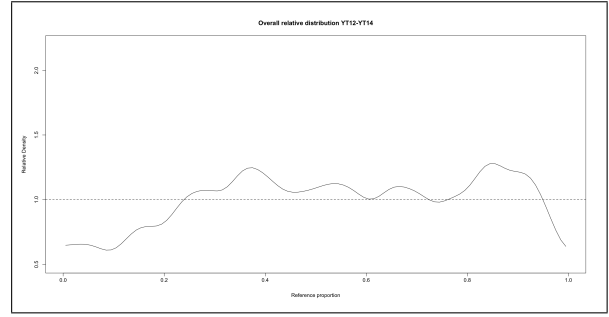


Figure 154: Overall relative distribution YT12-YT14

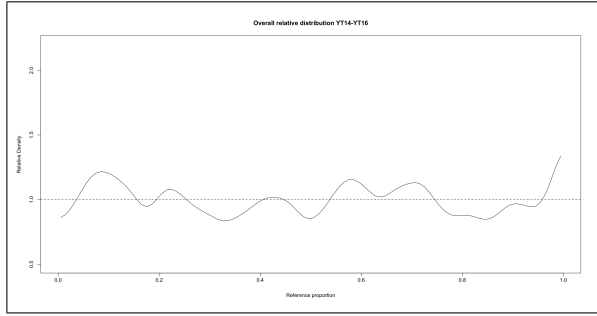


Figure 155: Overall relative distribution YT14-YT16

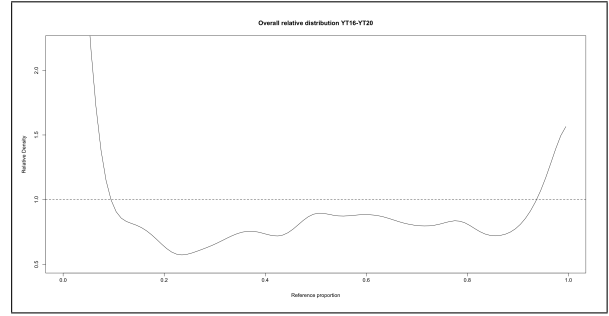


Figure 156: Overall relative distribution YT16-YT20

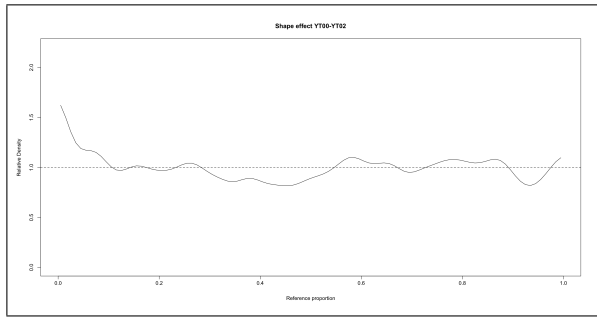


Figure 157: Shape effect YT00-YT02

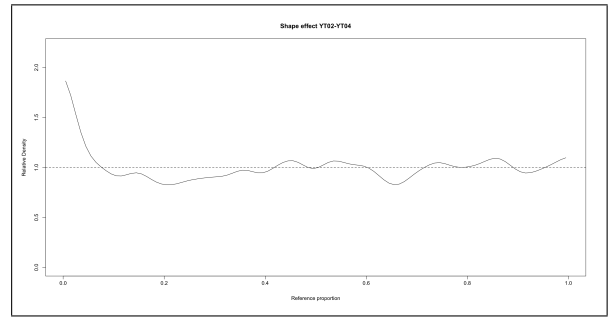


Figure 158: Shape effect YT02-YT04

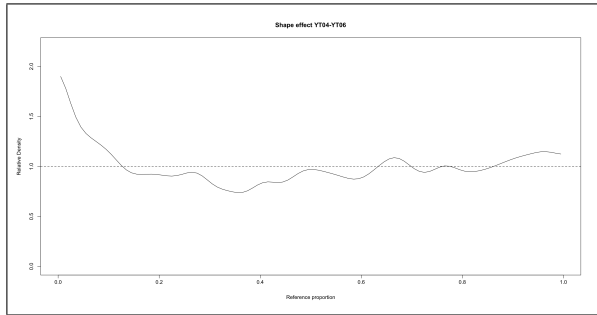


Figure 159: Shape effect YT04-YT06

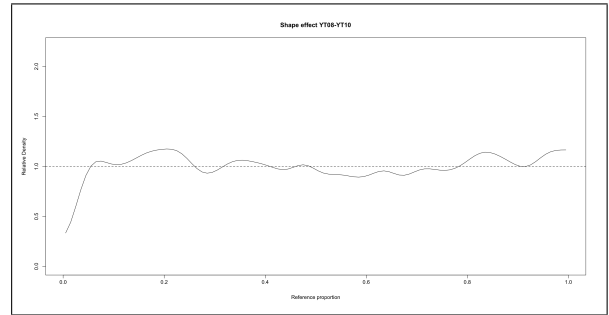


Figure 160: Shape effect YT08-YT10

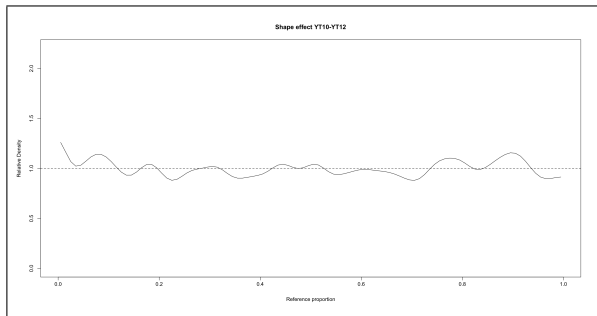


Figure 161: Shape effect YT10-YT12

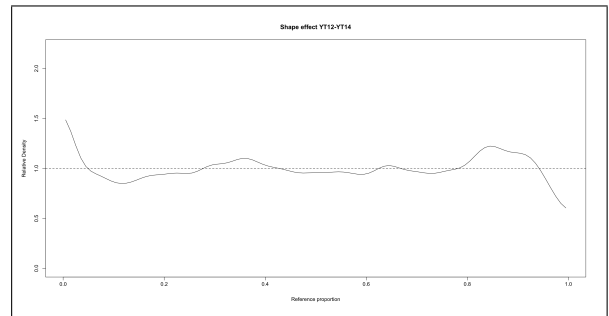


Figure 162: Shape effect YT12-YT14